Lab: Linear Regression with Kaggle House Prices Dataset

Estimated duration: 2–3 hours
Level: Intermediate
Dataset: House Prices - Advanced Regression Techniques (Kaggle)

Objectives

- Apply Linear Regression to real-world data.
- Explore, clean, and preprocess the dataset.
- Build a predictive model for house prices.
- Evaluate model performance using statistical metrics.
- Interpret regression coefficients and model goodness-of-fit.

1. Step 1: Data Preparation

Tasks

1. Download the `train.csv` file from Kaggle.
2. Load the dataset and display basic information: number of rows/columns, data types.
3. Check for missing data.

Discussion Questions

- How many columns have missing values?
- What strategy did you choose to handle missing data? Why?

2. Step 2: Exploratory Data Analysis

Tasks

1. Select 3–4 numerical features likely related to `SalePrice`, such as `GrLivArea`, `OverallQual`, `YearBuilt`, `GarageArea`.

2. Create scatter plots to visualize the relationship between each feature and `SalePrice`.
3. Calculate the Pearson correlation between selected features and `SalePrice`.

Discussion Questions

- Which variable has the strongest correlation with `SalePrice`? Is it positive or negative?
- Do you see any signs of non-linear relationships?

3. Step 3: Build the Linear Regression Model

Tasks

1. Split the data into training and testing sets (e.g., 80/20).
2. Train a Linear Regression model using selected features.
3. Display the model's intercept and coefficients.

Discussion Questions

- Write down the regression equation you obtained.
- How much does `SalePrice` increase/decrease for one unit change in `GrLivArea`?
- Are there any features with negative coefficients? What do they imply?

4. Step 4: Evaluate the Model

Tasks

1. Predict `SalePrice` on the test set.
2. Report the following metrics: Mean Squared Error (MSE) and R-squared (R^2).
3. Plot a chart comparing actual vs. predicted house prices.

Discussion Questions

- What is the R^2 value? What does it indicate about your model's performance?
- In which price range (low/high) does the model perform poorly?

5. Step 5: Extension (Optional)

Suggested Extensions

- Add more features such as `TotalBsmtSF`, `GarageArea`, and `LotArea`. Compare performance.
- Normalize or standardize the data before fitting the model. Observe the impact on coefficients.
- Try using interaction terms (e.g., `GrLivArea * OverallQual`) to see if it improves performance.

Discussion Questions

- Does adding more features improve the model?
- Are there any signs of **multicollinearity** in your model?

Report Submission

Each group must submit a report (PDF or Jupyter Notebook) that includes:

1. Description of selected features and preprocessing steps.
2. Visualizations, model results, and evaluation metrics.
3. Answers to all discussion questions.
4. A short analysis of the model's strengths and limitations.